

Justin Rhee

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EDUCATION

University of Colorado Boulder

Bachelor of Science in Computer Science, Minor in Business | GPA: 3.825

Boulder, CO

August 2024 – Present

EXPERIENCE

Software Development Intern

June 2025 – August 2025

Ecliptic Enterprises

Pasadena, CA

- Delivered automation tools, including laser tracking and document processing systems, adopted by engineering teams to improve testing precision and streamline documentation workflows
- Accelerated product validation by developing and scaling a Raspberry Pi-based prototype coelostat into an operational laser tracking system integrated into optical testing workflows
- Presented both projects to managers and clients, demonstrating performance outcomes and operational impact

Course Assistant - Data Science with Probability and Statistics

August 2025 – Present

University of Colorado Boulder

Boulder, CO

- Held triweekly office hours to support a class of 160+ students in probability, statistics, and data science
- Clarified statistical concepts, debugging, and programming practices by answering technical questions on Piazza

Business/IT Intern

May 2024 – August 2024

Kids in Motion Pediatric Therapy

Torrance, CA

- Automated data transfer between the old and new Electronics Medical Records systems by applying technical Python programming skills, saving 200+ hours of staff time
- Tracked, managed, and optimized staff case-load and productivity in collaboration with the Clinic Director

PROJECTS

Dual-Axis Laser-Tracking Coelostat | *Python, PySide, OpenCV, Threading*

July 2025

- Designed a dual-axis motor control system for 360° laser tracking with real-time image feedback, averaging a 9 Hz command rate dynamically timed to motor step completion based on step size, speed, and acceleration
- Built a PySide GUI with dual-camera feeds, centroid overlay and status indicator, and real-time data visualization
- Achieved sub-degree tracking stability with $\pm 0.05^\circ$ precision and $\pm 0.1^\circ$ tolerance for consistently smooth motion
- Implemented PID control and low-pass filtering to stabilize motor response and enhance tracking accuracy
- Applied HSV segmentation, contour analysis, and morphological filtering to track centroids under variable lighting
- Used multithreading for dual-camera coordination, processing, motion, and data tasks, reducing feedback latency

Document Automation System | *Python, Tkinter, importlib*

August 2025

- Engineered a backend processor for placeholder replacement and structured logging across 100+ Office templates using control-file mapping, recursive format traversal, and version-safe output, cutting manual entry by $\approx 70\%$
- Built a multi-stage Tkinter GUI for folder navigation, placeholder entry, and batch processing of Office templates
- Implemented importlib-based auto-installer for dependency setup across new systems
- Designed collapsible folder trees and summary views showing selected files and results for clearer navigation

RELEVANT COURSEWORK

Introduction to Computer Science | *CSCI 1300*

Summer 2024

Data Structures | *CSCI 2270*

Fall 2024

Computer Systems | *CSCI 2400*

Spring 2024

Introduction to Data Science with Probability/Statistics | *CSCI 3022*

Spring 2024

Linear Algebra with Computer Science Applications | *CSCI 2820*

Spring 2024

Algorithms | *CSCI 3104*

Fall 2025

Introduction to Artificial Intelligence | *CSCI 3202*

Fall 2025

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS

Developer Tools: VS Code, GitHub, Git, Jupyter Notebook, pip, Visual Studio, Docker

Libraries: PySide, OpenCV, Threading, importlib, pandas, NumPy, Matplotlib